

TR40 LiDAR Analysis Suite

Elastic Mie LiDAR · 532 nm · NARIT Chiang Mai · 5-step Processing Pipeline

Abstract

This document summarises the TR40 ground-based atmospheric LiDAR analysis suite developed for the National Astronomical Research Institute of Thailand (NARIT), Chiang Mai. The suite converts raw photon-counting / analog detector files into research-grade products: Normalized Relative Backscatter (NRB), Aerosol Layer Top (ALT) via the Haar Wavelet Covariance Transform, Range-Time-Intensity (RTI) visualisation, and Fernald / Klett inversion for aerosol backscatter, extinction, optical depth and mass concentration. A 5-page summary is provided here; the full code base (~9000 lines of Python) is open source.

Key capabilities

Step	Page	What it does
Step 1	MPL rmin-rmax	Build the search-window table from Mini-MPL CSV files (sun photometer + MPL PBL).
Step 2	NRB Profile	TR40 .dat -> dead-time correction -> cosine glue -> BG / overlap / afterpulse -> R^2 -> normalize.
Step 3	ALT Detection	FFT low-pass -> HWCT Haar step -> most-negative W peak inside guided / profile / dual window.
Step 4	RTI Visualizer	Prototype vs Mini-MPL heatmap, ALT overlay (white line + black halo), compare panel, cloud layers.
Step 5	Fernald / Klett	Aerosol backscatter, extinction, AOD, mass concentration ($\mu\text{g}/\text{m}^3$) + AERONET AOD validation.

Generated 10 June 2026 · Source: github.com/alongkorn271-lgtm/tr40-lidar-analysis

Hardware specification

Component	Specification
Telescope	Celestron EdgeHD 800 · D = 203 mm, f = 2032 mm, F/10
Field stop	Diameter 8.0 mm -> Receiver FOV = 3.94 mrad (full)
Laser	Quantel VIRON @ 532 nm · D4σ = 3.8 mm · divergence 1.5 mrad (full)
Detector	Licel PM-HV + Hamamatsu R9880U PMT (8 mm cathode)
Geometry	Biaxial · d_perp = 148.12 mm · bin spacing 3.75 m (25 ns)
Overlap O(R)	Rice CDF (Kumar 2013): O=0.5 at 79 m, O=0.99 at 142 m

Processing pipeline

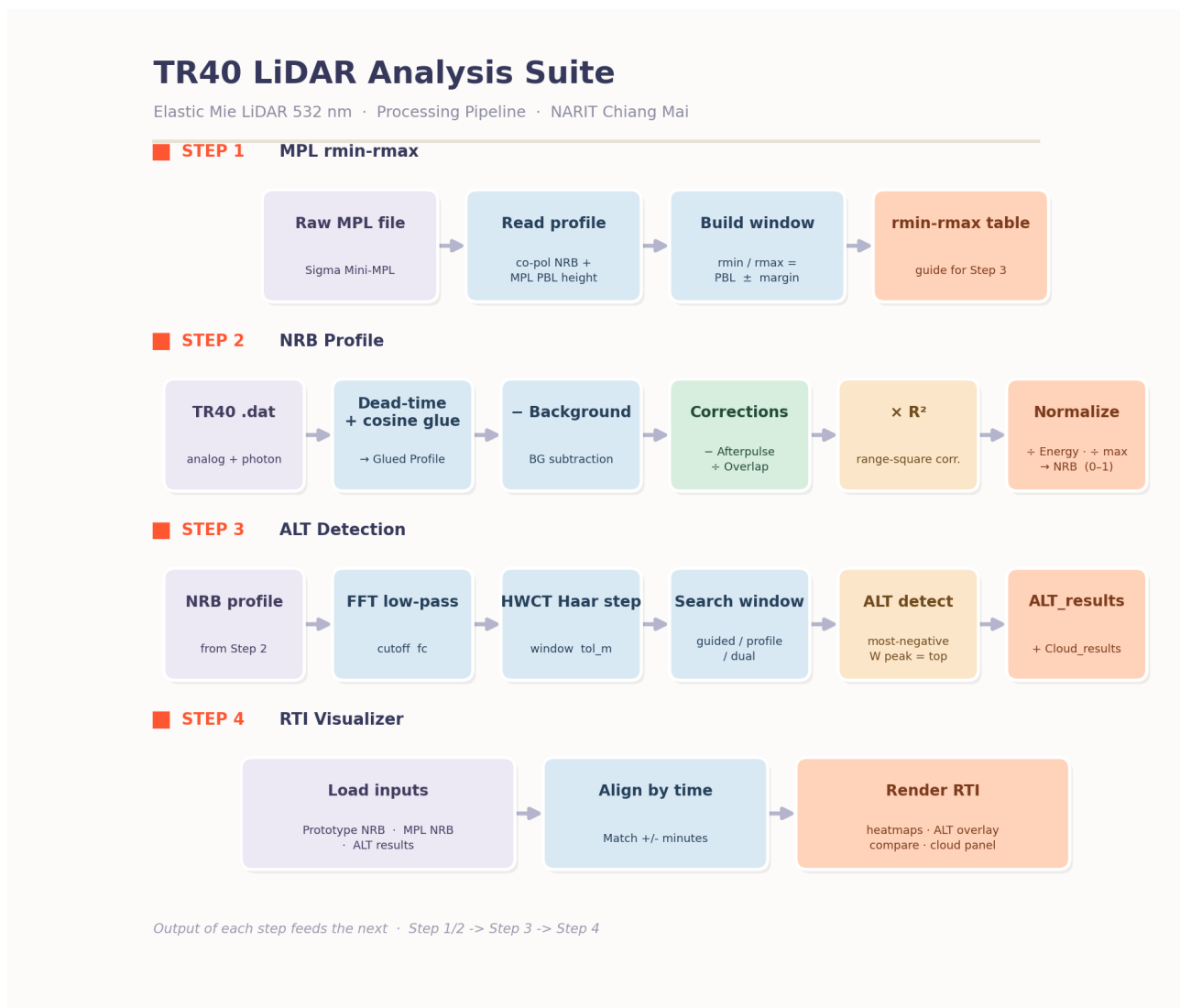


Figure 1 — Five-step processing pipeline. Output of each step feeds the next.

Step 2 — NRB Profile construction

Normalized Relative Backscatter follows the SigmaMPL convention. Raw analog + photon-counting profiles are merged using cosine-tapered analog–photon glue after dead-time correction; background, afterpulse, and the geometric overlap function are subtracted / divided out; the result is range-squared corrected, energy-normalised, and finally normalised to its own maximum for plotting.

$$\text{NRB} = (\text{Raw} \cdot \text{DT} - \text{BG} - \text{A}(\text{R})) / \text{O}(\text{R}) \times \text{R}^2 / \text{E_pulse} , \text{ then } \div \text{max}$$

Order of operations: dead-time → glue → −BG → −afterpulse → ÷overlap → ×R² → ÷energy → ÷max.

Overlap function O(R)

O(R) is computed analytically with the Gaussian-beam + Rice CDF model (Kumar & Rocadenbosch 2013) using the NARIT hardware spec above. For the TR40 setup, full overlap is essentially reached above ~150 m — well below the typical boundary-layer top.

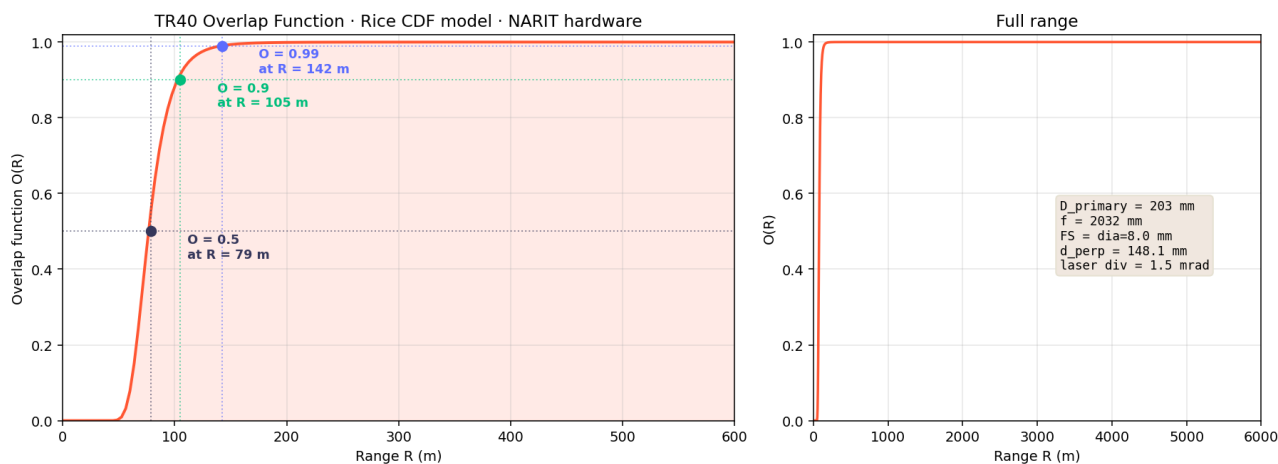


Figure 2 — TR40 overlap function (left: zoom 0–600 m, right: full range).

Threshold	Range R (m)	Notes
$O \geq 0.10$	63.75	lowest trusted range (with O _{min} = 0.1 mask)
$O \geq 0.50$	78.75	half overlap
$O \geq 0.90$	105.00	near full overlap
$O \geq 0.99$	142.50	full overlap (effective)

Step 3 — Aerosol Layer Top (ALT) detection

Each profile is FFT low-pass filtered then convolved with a Haar wavelet $W = \text{mean}(\text{upper}) - \text{mean}(\text{lower})$. The aerosol layer top is identified as the most-negative W peak inside the search window (falling edge of backscatter, i.e. layer top). Engine version 3 (May 2026) corrected an earlier preference for positive peaks that mistook the BL base for its top.

Mode	Search window	Behaviour
mpl_guided	[rmin, rmax] from MPL	best accuracy; MAE $\approx$ 95–115 m vs MPL
nrb_profile	[profile_rmin, profile_rmax]	MPL-free; needs sharp BL top + tuned tol_m
dual	both (default)	guided result selected first, profile as fallback

Step 5 — Fernald / Klett aerosol inversion

Backward Fernald (1984) inversion separates aerosol and molecular contributions. Self-calibrating via the Rayleigh boundary condition $\beta_{\text{total}}(R_{\text{ref}}) = \beta_{\text{mol}}(R_{\text{ref}})$, so no separate lidar constant is needed and the NRB normalisations cancel algebraically. Outputs are written to a separate Fernald-<date>.xlsx workbook with sheets for β_{aer} , α_{aer} , AOD, mass concentration, and (optionally) AERONET comparison.

Recommended parameters · Chiang Mai burning season

Parameter	Value	Reason
Method	Fernald 1984	publication standard
R_ref (m)	6000–7000	above smoke layer (can reach 4–5 km)
S_a (sr)	70	biomass smoke 532 nm (Janjai 2009, Pani 2019)
Atmosphere	US Standard Atm	Chiang Mai $\sim$ 300 m ASL — error $<5\%$
σ_{ext} (m ² /g)	4.0	biomass smoke (Reid 2005, Hand & Malm 2007)
HWCT tol_m (m)	250	matches TR40 broad BL transition ($\sim$ 400–800 m wide)
Time-match (min)	15	AERONET cadence $\sim$ 15 min

Validation against AERONET (Chiang_Mai_Met_Sta)

Fernald-derived AOD at 532 nm from the TR40 LiDAR is compared with co-located AERONET (Aerosol Robotic Network, NASA) direct-sun retrievals. AERONET AOD at 532 nm is obtained directly when available, otherwise Ångström-interpolated from the 500 nm and 440 nm channels. AERONET operates only during daylight (sun photometer), so direct time-matching requires twilight overlap windows.

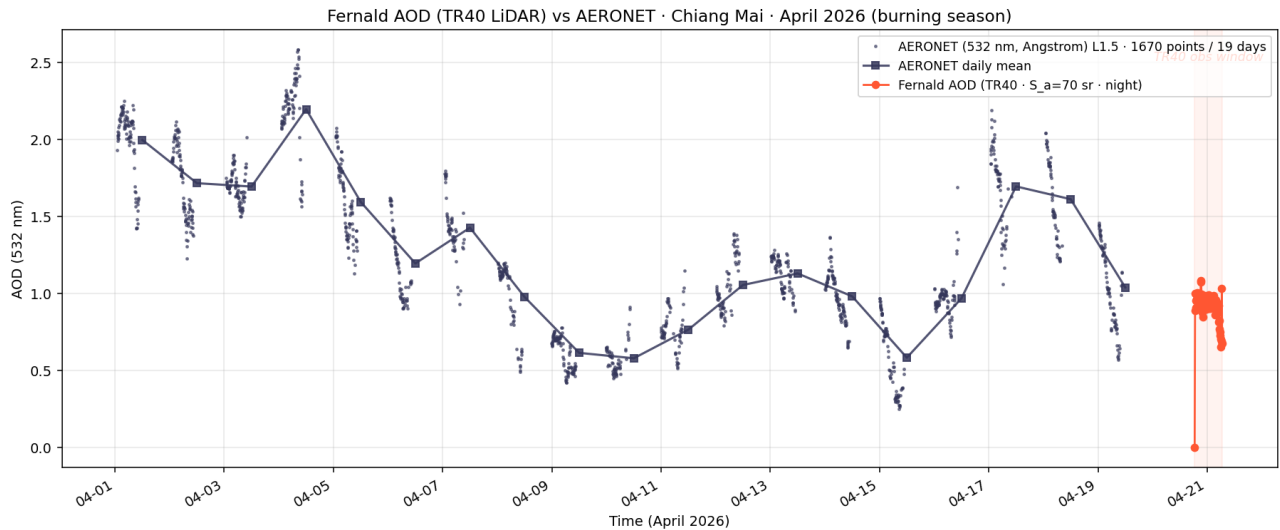


Figure 3 — April 2026 burning season. AERONET daily mean AOD (navy line) with raw 10-min retrievals (dots). The TR40 LiDAR Fernald AOD for 20–21 April night (orange) sits within the same magnitude range as the surrounding sunny days.

Findings

- TR40 Fernald night-mean AOD = 0.91; AERONET 19 April day-mean = 1.04 — consistent within burning-haze variability.
- April 2026 AERONET daily-mean range 0.58–2.20 (peak 4 April at AOD 2.20). TR40 obs window catches the declining edge after 17–19 April peak.
- Direct time-matched validation impossible for this dataset: TR40 obs = night, AERONET 20–22 April = no records (cloud-screened to zero).

Limitations / open work

- Daytime TR40 NRB SNR is hardware-limited by solar background → daytime Fernald is qualitative only.
- Cross-polarization channel not yet installed → no δ -based aerosol typing or improved ALT.
- Recommended fix: add cross-pol + narrowband interference filter (1 nm @ 532 nm).
- For rigorous AERONET validation, schedule TR40 to capture twilight windows (05:30–07:00 and 17:00–18:30 local).

Output file structure

Step	File	Sheets
1	rmin-rmax-<date>.xlsx	rmin-rmax, copol_nrb_norm
2	NRB-<date>.xlsx	NRB profile, QC_params, Parameters
3	ALT_NRB-<date>.xlsx	Input_NRB_raw, Input_rmin-rmax, NRB_Denoised_FFT, ALT_results, Cloud_results, Summary, Parameters
4	(plots / PNG)	RTI heatmap, ALT overlay, ALT compare, cloud panel
5	Fernald-<date>.xlsx	Fernald_beta_aer, Fernald_alpha_aer, Fernald_AOD, Fernald_mass_aer, AERONET_comparison

Source code & data

Code: github.com/alongkorn271-lgtm/tr40-lidar-analysis

Run desktop GUI: `python main_v2.py`

Run web GUI: `streamlit run streamlit_app.py`

AERONET sample: MPL data/AERONET_Chiang_Mai_Met_Sta_2026-Jan-May_L1.5.lev15

References

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